

Autonomous Production Choke Controller for a Single Naturally Flowing Oil Well

Honeywell Hackathon Project Report

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Abstract

The oil and gas industry increasingly relies on intelligent automation to improve production efficiency while ensuring operational safety. One of the critical components in oil well production is the choke valve, which regulates the flow of hydrocarbons from the reservoir to the surface. Selecting an inappropriate choke opening can lead to reduced oil production, excessive pressure variations, and unsafe operating conditions.

This project presents a data-driven Autonomous Production Choke Controller for a naturally flowing oil well. The proposed solution utilizes historical operational data to analyze the relationship between choke position, oil production rate, and pressure measurements, including Wellhead Pressure (WHP), Flowline Pressure (FLP), and Bottom Hole Pressure (BHP). Exploratory Data Analysis (EDA) and step-test analysis are performed to understand system behaviour, followed by the development of machine learning-based predictive models using Linear Regression.

The trained models are integrated with a safety-aware decision-making algorithm that evaluates multiple choke positions, predicts their corresponding production and pressure values, verifies compliance with predefined safety constraints, and automatically selects the optimal choke opening that maximizes oil production. The proposed controller demonstrates how predictive analytics and intelligent control can enhance production efficiency while maintaining safe operating conditions. This solution provides a scalable foundation for future integration with real-time oilfield monitoring systems and advanced control strategies.

Introduction

Oil wells produce hydrocarbons under varying reservoir and surface operating conditions. The production choke valve plays a vital role in controlling the flow of fluids from the well by regulating the choke opening. Proper choke management is essential because it directly affects oil production, pressure stability, equipment safety, and overall operational efficiency.

Traditionally, choke settings are determined manually by experienced field operators or based on predefined operating procedures. Although effective, manual control can be time-consuming, may not respond quickly to changing well conditions, and often depends on operator expertise. As oilfield operations become increasingly automated, there is a growing need for intelligent systems capable of making optimal control decisions with minimal human intervention.

The objective of this project is to develop an Autonomous Production Choke Controller that automatically determines the most suitable choke opening based on predicted production and pressure values. Using historical operational data, machine learning techniques are employed to model the relationship between choke position and key production parameters. A safety-aware optimization algorithm then evaluates multiple choke candidates and selects the one that provides the highest oil production while ensuring that pressure values remain within acceptable operating limits.

This approach combines predictive analytics with intelligent decision-making to improve production performance, reduce operational risks, and support the transition toward smart and autonomous oilfield operations.

Problem Statement

Efficient production from naturally flowing oil wells requires continuous optimization of the choke valve position. Increasing the choke opening can improve oil production; however, excessive opening may result in unsafe operating pressures, equipment damage, production instability, or reduced well life. Conversely, conservative choke settings may ensure safe operation but lead to underutilization of the well's production potential.

The challenge is to develop an intelligent controller capable of automatically selecting the optimal choke opening under changing operating conditions. The controller must maximize oil production while ensuring that important operational parameters, including Wellhead Pressure (WHP), Flowline Pressure (FLP), and Bottom Hole Pressure (BHP), remain within predefined safety limits.

To address this challenge, this project proposes a machine learning-based autonomous choke controller that predicts production and pressure responses for different choke positions, evaluates multiple operating scenarios, applies safety constraints, and recommends the choke setting that provides the best balance between production optimization and operational safety. The proposed solution demonstrates the practical application of artificial intelligence and predictive modeling for intelligent oilfield production optimization.

Objectives

The primary objective of this project is to develop an intelligent and autonomous production choke controller that optimizes oil well performance while ensuring operational safety. The specific objectives are as follows:

- **Maximize Oil Production:** Develop a control strategy that identifies the optimal choke opening to achieve the highest possible oil production without compromising safety.
- **Maintain Safe Operating Pressure Limits:** Ensure that critical pressure parameters, namely Wellhead Pressure (WHP), Flowline Pressure (FLP), and Bottom Hole Pressure (BHP), remain within predefined safe operating limits during controller operation.
- **Develop Predictive Machine Learning Models:** Build data-driven predictive models capable of estimating oil production and pressure values for different choke openings using historical operational data.
- **Automate Choke Selection:** Design an autonomous decision-making algorithm that evaluates multiple choke positions and automatically recommends the most suitable choke setting based on production and safety criteria.
- **Validate Controller Performance:** Evaluate the effectiveness of the proposed controller through simulation under different operating scenarios and analyze its ability to optimize production while maintaining system stability.

Dataset Description

The project utilizes a simulated oil well production dataset provided for the Honeywell Hackathon. The dataset contains 120 observations, representing the operating conditions of a naturally flowing oil well over a period of time.

Each observation consists of the following variables:

Variable	Description
Time (hr)	Time at which the measurements were recorded.
Choke Percentage (%)	Percentage opening of the production choke valve.
Oil Rate (bbl/hr)	Oil production rate measured in barrels per hour.
Wellhead Pressure (WHP)	Pressure measured at the wellhead.
Flowline Pressure (FLP)	Pressure within the production flowline.
Bottom Hole Pressure (BHP)	Pressure at the bottom of the well near the reservoir.

The dataset serves as the foundation for the entire project. It is used to perform Exploratory Data Analysis (EDA), understand relationships among production variables, conduct step-test analysis, train machine learning models, and simulate the autonomous choke controller. Since the data represents realistic well behavior, it enables the development and evaluation of a predictive control strategy under various operating conditions.

Methodology

The implementation of the proposed autonomous production choke controller follows a systematic workflow consisting of data analysis, predictive modeling, and intelligent decision-making. The methodology adopted in this project is illustrated below:

Step 1: Data Loading and Preprocessing

The simulated dataset is imported into the Python environment using the Pandas library. The dataset is examined for missing values, data consistency, and statistical characteristics to ensure it is suitable for analysis and model development.

Step 2: Exploratory Data Analysis (EDA)

Exploratory Data Analysis is performed to understand the behavior of the oil well. Time-series plots, scatter plots, and summary statistics are used to visualize trends in oil production, choke position, and pressure variables.

Step 3: Correlation Analysis

A correlation heatmap is generated to examine the relationships among the numerical variables. This analysis helps identify the influence of choke opening on oil production and pressure measurements, providing insights for predictive modeling.

Step 4: Step-Test Analysis

The response of the oil well to changes in choke opening is analyzed using step-test analysis. This helps understand the dynamic relationship between choke adjustments and the corresponding changes in oil production and pressure values.

Step 5: Machine Learning Model Development

Linear Regression models are developed using the Scikit-learn library to predict oil production rate and pressure variables (WHP, FLP, and BHP). The models are trained using historical operational data and evaluated using performance metrics such as R^2 Score, Mean Absolute Error (MAE), and Root Mean Square Error (RMSE).

Step 6: Candidate Choke Generation

The controller generates a set of possible choke openings around the current operating point. These candidate values represent the possible actions available for optimizing oil production.

Step 7: Prediction of Operating Parameters

For each candidate choke opening, the trained machine learning models predict the expected oil production rate and pressure values. These predictions enable the controller to evaluate the consequences of each possible control action.

Step 8: Safety Constraint Validation

The predicted operating conditions are compared against predefined pressure limits. Any choke candidate that violates safety constraints is discarded, ensuring that only safe operating conditions are considered.

Step 9: Optimal Choke Selection

Among the safe candidate choke openings, the controller selects the one that provides the highest predicted oil production. The selected choke opening represents the optimal operating point for the current well conditions.

Step 10: Simulation and Performance Evaluation

The complete controller is simulated under different operating scenarios to evaluate its effectiveness. The simulation results are analyzed using dashboard visualizations, production trends, and pressure profiles to demonstrate the controller's ability to maximize oil production while maintaining safe operating conditions.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to gain a comprehensive understanding of the simulated oil well production dataset before developing the predictive models. This stage helped identify patterns, trends, and relationships among the operational variables while ensuring the quality and consistency of the data.

Initially, the dataset was inspected to verify its structure, data types, descriptive statistics, and the presence of missing values. Since no missing or inconsistent values were observed, the dataset was considered suitable for further analysis.

Several visualization techniques were employed to better understand the behavior of the production system:

- **Time-Series Analysis:** Time-series plots were generated for Choke Percentage, Oil Rate, Wellhead Pressure (WHP), Flowline Pressure (FLP), and Bottom Hole Pressure (BHP) to observe how these variables changed over time. These plots helped identify production trends and variations throughout the simulation period.
- **Scatter Plot Analysis:** Scatter plots were used to examine the relationship between choke opening and each operational parameter. These visualizations provided insights into how changes in choke position influenced oil production and pressure values.
- **Correlation Analysis:** A correlation heatmap was generated to measure the strength of relationships among all numerical variables. The heatmap helped identify positive and negative correlations between choke opening, oil production, and pressure measurements, providing valuable information for selecting features during model development.

The EDA results indicated that choke opening has a significant influence on oil production and pressure behavior, confirming that the available dataset is suitable for developing predictive machine learning models for autonomous choke control.

Step-Test Analysis

Step-test analysis was performed to understand the dynamic response of the oil well to changes in choke valve opening. In industrial process control, step tests are commonly used to study how a system reacts when the control input is changed by a fixed amount.

For this project, variations in the choke opening percentage were analyzed over time, and the corresponding responses in Oil Rate, Wellhead Pressure (WHP), Flowline Pressure (FLP), and Bottom Hole Pressure (BHP) were observed. The analysis helped determine the sensitivity of production and pressure variables to choke adjustments.

The step-test analysis provided several important insights:

- An increase in choke opening generally resulted in changes in oil production and pressure values.
- Pressure responses occurred simultaneously with production changes, indicating strong coupling between these variables.
- Different choke positions produced varying operating conditions, emphasizing the importance of selecting an optimal choke setting.
- The observed system behavior supported the development of a predictive controller capable of estimating future operating conditions before implementing control actions.

The knowledge obtained from the step-test analysis served as the basis for designing the autonomous choke controller, enabling it to make informed decisions regarding choke adjustments while maintaining operational safety.

Dynamic Model Development

A data-driven dynamic model was developed to predict the behavior of the oil well under different choke operating conditions. Since obtaining real-time physical models of oil wells can be computationally expensive and complex, machine learning techniques were employed to create accurate predictive models using historical operational data.

Linear Regression was selected as the prediction algorithm due to its simplicity, computational efficiency, and ability to model linear relationships between operational variables. Separate regression models were trained to predict the following output parameters:

- Oil Production Rate (Oil Rate)
- Wellhead Pressure (WHP)
- Flowline Pressure (FLP)
- Bottom Hole Pressure (BHP)

The dataset was divided into training and testing sets to evaluate the model's predictive capability. During training, the algorithm learned the relationship between choke opening and the corresponding production and pressure variables.

The performance of each model was evaluated using standard machine learning evaluation metrics:

- R^2 Score (Coefficient of Determination): Measures how well the model explains the variability in the target variable. Higher values indicate better prediction accuracy.
- Mean Absolute Error (MAE): Measures the average magnitude of prediction errors without considering their direction. Lower MAE values indicate better model performance.

- Root Mean Square Error (RMSE): Measures the average prediction error while giving greater importance to larger errors. Lower RMSE values indicate more accurate predictions.

The trained regression models formed the predictive core of the autonomous controller. For every candidate choke position generated by the controller, these models estimated the expected oil production and pressure values. The controller then evaluated these predictions against predefined safety constraints and selected the choke opening that maximized oil production while maintaining safe operating conditions.

The dynamic modeling approach successfully demonstrated that machine learning can effectively capture the operational behavior of the oil well and provide reliable predictions for intelligent production optimization.

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Autonomous Controller

The Autonomous Production Choke Controller forms the core component of the proposed solution. Its primary objective is to automatically determine the optimal choke valve opening that maximizes oil production while ensuring that all operating pressures remain within predefined safety limits.

The controller operates by first generating a range of candidate choke positions around the current operating point. Each candidate represents a possible control action that can be applied to the production choke valve. These candidate values are then evaluated using the trained machine learning models developed during the dynamic modeling phase.

For every candidate choke opening, the predictive models estimate the corresponding Oil Production Rate, Wellhead Pressure (WHP), Flowline Pressure (FLP), and Bottom Hole Pressure (BHP). These predicted values are subsequently verified against predefined safety constraints to ensure that none of the operating pressures exceed acceptable limits.

Any candidate choke position that violates the safety criteria is immediately discarded. Among the remaining safe candidates, the controller selects the choke position that provides the highest predicted oil production. This optimization process allows the controller to continuously recommend the most efficient operating point while maintaining operational reliability and equipment safety.

The autonomous controller eliminates the need for continuous manual intervention by automatically making informed decisions based on predictive analytics. This demonstrates the

practical application of artificial intelligence and machine learning in modern oilfield production optimization.

Results and Discussion

The proposed autonomous choke controller was successfully implemented and evaluated using the simulated oil well dataset. The controller demonstrated its ability to predict production and pressure responses accurately and recommend suitable choke openings under different operating conditions.

The machine learning models effectively captured the relationship between choke position and the operational parameters, enabling reliable prediction of Oil Rate, Wellhead Pressure (WHP), Flowline Pressure (FLP), and Bottom Hole Pressure (BHP). Performance metrics such as R^2 Score, Mean Absolute Error (MAE), and Root Mean Square Error (RMSE) confirmed that the developed models provided satisfactory prediction accuracy for controller implementation.

The autonomous controller generated multiple candidate choke positions, evaluated their predicted outcomes, applied safety constraints, and consistently selected the choke opening that maximized oil production without violating pressure limits.

Several visualization dashboards were developed to analyze controller performance. These dashboards included:

- Time-series plots showing variations in oil production and pressure over time.
- Scatter plots illustrating the relationship between choke opening and production variables.
- Correlation heatmaps highlighting dependencies among operational parameters.
- Controller simulation plots demonstrating the selected choke positions under different operating scenarios.

The simulation results indicate that the proposed controller is capable of maintaining safe operating conditions while improving production efficiency. Overall, the developed system successfully demonstrates the feasibility of using machine learning-based autonomous control for intelligent oilfield production optimization.

Challenges and Limitations

Although the proposed solution achieved encouraging results, several limitations should be considered.

The primary limitation is that the controller was developed using a simulated dataset rather than real industrial field data. Real production environments often contain measurement noise, sensor failures, changing reservoir characteristics, and varying operational conditions that are difficult to represent completely using simulated data.

Another limitation is the use of Linear Regression as the predictive model. While Linear Regression provides simplicity, fast computation, and interpretability, it may not fully capture complex nonlinear relationships between choke position and production parameters that commonly occur in real oil wells.

Additionally, the controller assumes fixed operating constraints and static system behavior. In practical applications, operating conditions may change continuously due to reservoir depletion, equipment aging, fluid composition changes, and environmental factors.

The current implementation also operates in an offline simulation environment. Real-time deployment would require continuous monitoring of sensor data, automatic model updates, communication with industrial control systems, and extensive field validation before practical implementation.

Despite these limitations, the project successfully demonstrates the feasibility of developing an intelligent autonomous choke controller using data-driven techniques.

Future Scope

The proposed autonomous production choke controller provides a strong foundation for future research and industrial implementation. Several enhancements can be incorporated to further improve its accuracy, robustness, and practical applicability.

One possible extension is the integration of the controller with real-time oilfield simulators and Supervisory Control and Data Acquisition (SCADA) systems. This would enable continuous monitoring of production data and automatic choke adjustment based on live operating conditions.

More advanced machine learning algorithms such as Random Forest, Gradient Boosting, XGBoost, Artificial Neural Networks (ANN), and Deep Learning models can be explored to improve prediction accuracy for complex nonlinear production systems.

Future work may also include the implementation of Model Predictive Control (MPC), which considers future system behavior while optimizing current control actions. MPC can provide smoother choke adjustments and better handling of process constraints.

Another promising direction is the application of Reinforcement Learning, where the controller learns optimal control policies through continuous interaction with the production environment instead of relying solely on historical data.

Additional future enhancements include:

- Integration with live sensor data and Industrial Internet of Things (IIoT) devices.
- Automatic retraining of machine learning models using newly acquired production data.
- Cloud-based deployment for remote monitoring and centralized control.
- Multi-well production optimization instead of single-well operation.
- Digital Twin implementation for real-time performance monitoring and predictive maintenance.

These improvements would further enhance the capability of autonomous production systems and support the development of fully intelligent oilfield operations.

Conclusion

This project successfully developed a data-driven Autonomous Production Choke Controller for a naturally flowing oil well using machine learning and safety-aware optimization techniques.

Historical production data was analyzed through Exploratory Data Analysis (EDA), correlation analysis, and step-test analysis to understand the relationships between choke position, oil production, and pressure variables. Linear Regression models were then developed to predict Oil Rate, Wellhead Pressure (WHP), Flowline Pressure (FLP), and Bottom Hole Pressure (BHP).

The predictive models were integrated with an autonomous decision-making algorithm that generated multiple candidate choke positions, evaluated their predicted outcomes, applied operational safety constraints, and selected the choke opening that maximized oil production while maintaining safe pressure limits.

Simulation results demonstrated that the proposed controller can effectively optimize production without compromising operational safety. The developed solution highlights the potential of machine learning and intelligent control techniques in modern oilfield automation.

Overall, the project demonstrates a practical, scalable, and cost-effective approach toward autonomous production optimization and provides a strong foundation for future deployment in real-time industrial environments.

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